



- 100% three-phase unbalanced output, with each phase capable of delivering up to 50% of the rated power.
- Ac coupling to retrofit existing solar system
- Max. 6 pcs parallel for ongrid and off-grid operation; Support multiple batteries parallel
- Max. charging/discharging current of 240A
- 6 time periods for battery charging/discharging
- Support storing energy from diesel generator
- 48V low voltage battery, transformer isolation design

Three Phase Hybrid Inverter

TLH-HBY Series

Model	HYB-06KTLH-01-TVN	HYB-08KTLH-01-TVN	HYB-10KTLH-01-TVN	HYB-12KTLH-01-TVN	HYB-15KTLH-01-TVN
Battery Input Parameters					
Battery Type	Lithium or Lead-acid Battery				
Rated Battery Voltage(V)	48V				
Maximum Charging Voltage (V)	≤60V(configurable)				
Maximum Charging/Discharging Current (A)	120A	160A	210A	250A	280A
PV Input Parameters					
Maximum DC Input Power(kW)	10kW	12kW	20kW	24kW	30kW
Maximum DC Input Voltage (V)	1000				
MPPT Voltage Range (V)	200-800V				
Start-up Voltage(V)	150V				
Maximum Input Current(A)	20A/20A		36A/18A		
Number of MPPT Trackers	2				
AC Output Parameters(Grid-tied)					
Maximum Apparent Output Power(VA)	6600	8800	11000	13200	16500
Rated Output Voltage(V)	380/400V				
Rated Output Frequency(Hz)	50/60Hz				
Maximum Output Current (A)	10A	13.4A	16.7A	20A	25A
Output Power Factor	1(0.8 leading-0.8 lagging)				
AC Output Parameters(Off-grid)					
Rated Output Power(W)	6000	8000	10000	12000	15000
Peak Output Power(W)	>200%, 15sec				
Rated Output Voltage (V)	380/400V				
Rated Output Frequency(Hz)	50/60Hz				
Rated Output Current(A)	9.1A	12.1A	15.2A	18.2A	22.8A
Efficiency					
Max.Efficiency(PV)	98%				
Max.Efficiency(Battery)	94.5%				
European Efficiency	97.5%				
Protection					
PV Reverse Polarity Protection	Integrated				
PV Insulation Resistance Detection	Integrated				
Residual Current Detection	Integrated				
Output Overcurrent Protection	Integrated				
Output Short-circuit Protection	Integrated				
Output Overvoltage Protection	Integrated				
General Parameters					
Operating Temperature(°C)	-25~60°C(derating above 45°C)				
Storage Temperature(°C)	-30°C~65°C				
Relative Humidity	0~95%				
Operating Altitude(m)	≤4000m(derating above 2000m)				
Cooling Method	Intelligent Forced Air Cooling				
Noise(dB)	<55dB				
Weight(kg)	38kg				
Dimensions(W×H×Dmm)	475*683*256mm				
IP Rating	IP66				
Topology	High-frequency Isolation(Battery Side)				
Safety/EMC Certification	IEC62109,IEC61000,ISI6169&ISI6221(BIS)				
Grid Compliance	IEC61727,EN50549-1,VDE-4105,NRS-097-2-1,OVE-Richtlinie R25,UNE217001/2,Ordinace No.140				
Other Certifications	IEC61683,IEC62116,EN50530,IEC60068				